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Key Points:
- We evaluate biases of two common approximations to Wet-bulb globe temperature (WBGT) in comparison with explicitly calculated WBGT and consider labor implications
- Simplified WBGT generally overestimates heat stress and resulting labor loss, whereas environmental stress index is less biased. Both methods underestimate peak heat stress
- Both methods are systematically biased. We offer a computationally efficient Python implementation to encourage accurate WBGT calculations

Supporting Information:
Supporting Information may be found in the online version of this article.

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Explicit Calculations of Wet-Bulb Globe Temperature Compared With Approximations and Why It Matters for Labor Productivity

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Abstract Wet-bulb globe temperature (WBGT) is a widely applied heat stress index. However, most applications of WBGT within the heat stress impact literature that do not use WBGT at all, but use one of the ad hoc approximations, typically the simplified WBGT (sWBGT) or the environmental stress index (ESI). Surprisingly, little is known about how well these approximations work for the global climate and climate change settings that they are being applied to. Here, we assess the bias distribution as a function of temperature, humidity, wind speed, and radiative conditions of both sWBGT and ESI relative to a well-validated, explicit physical model for WBGT developed by Liljegren, within an idealized context and the more realistic setting of ERA5 reanalysis data. sWBGT greatly overestimates heat stress in hot-humid areas. ESI has much smaller biases in the range of standard climatological conditions. Over subtropical dry regions, both metrics can substantially underestimate extreme heat. We show systematic overestimation of labor loss by sWBGT over much of the world today. We recommend discontinuing the use of sWBGT. ESI may be acceptable for assessing average heat stress or integrated impact over a long period like a year, but less suitable for health applications, extreme heat stress analysis, or as an operational index for heat warning, heatwave forecasting, or guiding activity modification at the workplace. Nevertheless, Liljegren's approach should be preferred over these ad hoc approximations and we provide a fast Python implementation to encourage its widespread use.

Plain Language Summary Wet-bulb globe temperature (WBGT) is a widely applied heat stress index. However, most applications of WBGT within the climate change heat stress impact literature that do not use WBGT at all but use one of the ad hoc approximations, typically the simplified WBGT (sWBGT) or sometimes the environmental stress index (ESI). But we know little about how well these approximations work for measuring heat stress. Here, we evaluate the performance of sWBGT and ESI against a well-validated, explicit physical model of WBGT. sWBGT greatly overestimates heat stress under a hot, humid climate. ESI performs much better in measuring average heat stress. But they both may seriously underestimate severe heat stress, especially in hot, dry regions. Our results suggest that previous studies using sWBGT tend to dramatically overestimate heat stress and its economic and health implications, which may further misinform policymaking. We recommend discontinuing the use of sWBGT. ESI may be acceptable for assessing average heat stress, but less suitable for the warning or forecasting of extreme heat, or providing guidance for dealing with workplace heat stress. Nevertheless, the well-validated physical model of WBGT should be preferred over these approximations and we provide a Python implementation to encourage its widespread use.

1. Introduction

Heat stress has caused more deaths than any other extreme weather event (Barriopedro et al., 2011; Buzan et al., 2015) and is recognized to have broad social and economic impacts such as heat-related illness (Barriopedro et al., 2011; Ebi et al., 2021; Mora et al., 2017), conflict (Burke et al., 2009; Schleussner et al., 2016), crime (Shen et al., 2020), electricity demand (Maia-Silva et al., 2020), and labor productivity reduction (Dunne et al., 2013; Hsiang et al., 2017; Kjellstrom et al., 2016; Masuda et al., 2021; Orlov et al., 2020). Heat stress will become an even bigger threat in the future as the world warms (Diffenbaugh & Giorgi, 2012; C. Li, Sun, et al., 2020; D. Li, Yuan, & Kopp, 2020; Meehl & Tebaldi, 2004; Sherwood & Huber, 2010; Willett & Sherwood, 2010).

As well reviewed elsewhere, many heat stress metrics have been developed (De Freitas & Grigorieva, 2014; Epstein & Moran, 2006; Havenith & Fiala, 2015). Among these, the wet-bulb globe temperature (WBGT) in its various forms is arguably the most commonly used one, enjoying the advantages of a simple physical interpretation, covering all four ambient factors (temperature, humidity, wind, and radiation) contributing to heat stress,

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and having well-established safety thresholds to guide activity modification within the military (Army, 2003), occupational (ACGIH, 2017; ISO, 2017; NIOSH, 2016), and athletic settings (ACSM, 1984). WBGT is constructed as a linear combination of natural wet-bulb temperature (T_w), black globe temperature (T_g), and dry bulb temperature (T_a): WBGT = $0.7T_w + 0.2T_g + 0.1T_a$ (Yaglou & Minard, 1957). WBGT also has limitations such as underestimation of heat strain in case of low air movement and high humidity (Budd, 2008; Havenith & Fiala, 2015) and the difficulty in reflecting detrimental effects of wind under extremely hot, dry conditions (Foster et al., 2021a, 2021b). There are also other heat stress indices in use, such as the Universal Thermal Climate Index, which may be an improvement (Bröde et al., 2013; Foster et al., 2021a, 2021b; Havenith & Fiala, 2015); but here, we focus on WBGT because of its widespread use within climate change and heat stress literature.

The measurement of WBGT requires costly instruments and time-consuming attention by experienced operators, which prevents WBGT to become a routine meteorological measurement at weather stations. As a result, several approaches have been developed to approximate WBGT, with the simplified WBGT (sWBGT; ABM, 2010) and the environmental stress index (ESI) (D. Moran et al., 2001; D. Moran, Pandolf, Laor, et al., 2003) being representative of many similar ad hoc approaches.

sWBGT (ABM, 2010) is an approximate form requiring only temperature and humidity and explicitly assuming fixed moderately high solar radiation and low wind speeds, which implies potential positive or negative biases when these assumptions are not met. sWBGT has been widely used because of its simplicity for assessing heat stress and the implication on athletes and labor (Altinsoy & Yildirim, 2014; Cooper et al., 2016; Kakamu et al., 2017; Kjellstrom et al., 2009; Lee & Min, 2018; Liu, 2020; Smith et al., 2018; Willett & Sherwood, 2010; Zhu et al., 2021). ESI was constructed via a multiple regression with WBGT being the dependent variable and temperature, humidity, solar radiation, and their interaction terms being independent variables (D. Moran et al., 2001). ESI was validated across different climate regimes over Israel and New Zealand based on large databases (D. Moran, Pandolf, Shapiro, et al., 2003; D. Moran, Pandolf, Heled, & Gonzalez, 2004; D. S. Moran, Pandolf, Vitalis, et al., 2004; D. S. Moran et al., 2005). Although a high correlation (>0.9) between WBGT and ESI was achieved, the residual errors can be up to ±2°C, and it may be the critical situations (such as extreme heat stress) where ESI substantially under- or overestimates WBGT (Havenith & Fiala, 2015).

Outside of the limited conditions for which these approximate forms were developed, little is known about how well these approximations work for the global climate and climate change settings that they are being applied to. Although a few studies had quantified biases of sWBGT or ESI based on local meteorological measurements (Grundstein & Cooper, 2018; D. Moran, Pandolf, Heled, & Gonzalez, 2004; D. S. Moran, Pandolf, Vitalis, et al., 2004; D. S. Moran et al., 2005), the results are not readily transferable to other regions with different climate conditions. A recent study employed both sWBGT and ESI to assess labor reduction due to intensifying heat stress and found vast differences between the two metrics (De Lima et al., 2021). However, it is not clear which one is more close to reality. Given the expected biases of both metrics and their large discrepancies in indicating labor loss, it is necessary to assess the magnitude of these biases and the consequent influences on heat stress impact assessment, which is crucial for determining the suitability of each metric under certain application scenarios.

In addition, there have been studies using the psychrometric wet-bulb temperature (T_{pwb}) and air temperature to replace natural wet-bulb temperature and black globe temperature leading to the following formula: WBGT = $0.7*T_{pwb} + 0.3*T_a$ (Dunne et al., 2013; Knutson & Ploshay, 2016; Li et al., 2017; C. Li, Sun, et al., 2020; D. Li, Yuan, & Kopp, 2020; Newth & Gunasekera, 2018; Schwingshackl et al., 2021). This simplified form neglects the effects of solar radiation and wind speed making it only apply to indoor or well-shaded thermal conditions. Some other studies apply indoor or shaded WBGT ($WBGT_{id}$) approximations to outdoors by adding a constant factor (e.g., 3°C) to represent the effects of solar radiation (Dasgupta et al., 2021; Kjellstrom et al., 2014; Szewczyk et al., 2021), the validity of which has not been tested.

Aside from the simple approximations of WBGT described above, physical models on the energy balance of WBGT sensors have also been developed, which enable a direct simulation of WBGT measurements from weather station observations or climate model output (Bernard & Pourmoghani, 1999; Dernedde & Gilbert, 1991; Gaspar & Quintela, 2009; Hunter & Minyard, 1999; Liljegren et al., 2008). Among these physical models, the one developed by Liljegren et al. (2008) is highly sophisticated being well calibrated and validated (with an RMS difference of less than 1°C; Lemke & Kjellstrom, 2012; Liljegren et al., 2008). However, Liljegren's approach has seen limited applications (Casanueva et al., 2020; Jacobs et al., 2019; Orlov et al., 2019; Takakura et al., 2017, 2018) potentially because it is complex and computationally intensive. Moreover, Liljegren's code was written in C and FORTRAN languages, which may be not familiar to most end-users. To resolve this issue, we rewrote the code in Cython, which is fast, easy to use in Python, and scales well for large data set such as climate model output.

Here, we treat Liljegren's model as ground truth and explore the bias distributions of sWBGT and ESI within both an idealized context and the more realistic setting of ERA5 reanalysis data. We also briefly assessed the validity of using WBGT_{id} approximations to represent outdoors by adding a constant correction factor.

The paper is structured as follows. Section 2 introduces more details on the metrics and Liljegren's model, as well as data sources and analysis methods. Section 3 presents bias quantification results including, first, the bias distribution within an idealized context as a function of temperature, humidity, wind speed, and radiative conditions, and second, the error structure introduced within ERA5 reanalysis data. In Section 4, the potential consequences of these biases are examined through an example application of labor productivity estimation. Section 5 discusses the implication of our results. Section 6 concludes by highlighting the main findings and providing recommendations.

2. Data and Methods

2.1. sWBGT, ESI, and Liljegren's Model

Here, we present the formulas of sWBGT, ESI, and Liljegren's model. Parameter definitions and their units within all equations are summarized in the list of notations. sWBGT was developed for heat stress assessment in sports medicine (ACSM, 1984) and formulated as shown in Equation 1 within which T_a and e_a represent ambient air temperature and vapor pressure, respectively. sWBGT includes neither radiation nor wind speed two of the determining aspects of heat stress that are captured in WBGT.

$$sWBGT = 0.567\,(T_a - 273.15) + 0.393 e_a + 3.94 \tag{1}$$

ESI was designed as an approximation to WBGT via a multiple regression model including T_a, relative humidity (RH), surface downward solar radiation (S_{down}), and their interaction terms as independent variables (Equation 2) (D. Moran et al., 2001; D. Moran, Pandolf, Shapiro, et al., 2003). ESI excludes wind speed and thermal radiation.

$$ESI = 0.62\,(T_a - 273.15) - 0.7RH + 0.002 S_{down} + 0.43\,(T_a - 273.15)\cdot RH - 0.078(0.1 + S_{down})^{-1} \tag{2}$$

liljegren's model is physically based relying on fundamental principles of heat and mass transfer. It performs energy budget analyses on both natural wet-bulb and black globe sensors, which boil down to two separate equations for T_w (Equation 3) and T_g (Equation 5) (Liljegren et al., 2008) that need to be solved by iteration:

$$T_w = T_a - \frac{\Delta H}{c_p}\frac{M_{H2O}}{M_{Air}}\left(\frac{\Pr}{Sc}\right)^{0.56}\left(\frac{e_w - e_a}{P - e_w}\right) + \frac{\Delta F_{net}}{Ah} \tag{3}$$

where c_p is the specific heat of dry air at constant pressure; P and e_w are, respectively, surface pressure and vapor pressure at the surface of the wick; h is a convective heat transfer coefficient, which is a function of air temperature, pressure, wind speed, and the shape of T_w or T_g sensors; ΔF_{net} refers to net radiative gain by the wick:

$$\begin{aligned}\Delta F_{net} = \frac{1}{2}\pi DL\epsilon_w\,(L_{down} + L_{up}) - \pi DL\sigma\epsilon_w T_w^4 + \left(\pi DL + \frac{\pi D^2}{4}\right)(1-\alpha_w)\,(1 - f_{dir})\,S_{down} \\ + \left(DL\sin\theta + \frac{\pi D^2}{4}\cos\theta\right)(1-\alpha_w)\,f_{dir}\frac{S_{down}}{\cos\theta} + \pi DL\,(1-\alpha_w)\,S_{up}\end{aligned} \tag{4}$$

where S_{down}, S_{up}, L_{down}, and L_{up} denote surface downward and upwelling solar and long-wave radiation, respectively; f_{dir} represents the fraction of the total horizontal solar irradiance due to the direct beam of the sun; σ is the Stefan-Boltzmann constant; ϵ_w and α_w are, respectively, the emissivity and albedo of the wick. The energy budget analysis on the black globe sensor leads to the equation below for T_g:

$$T_g^4 = \frac{L_{down} + L_{up}}{2\sigma} - \frac{h\,(T_g - T_a)}{\epsilon_g\sigma} + \frac{S_{down}\,(1-\alpha_g)}{2\epsilon_g\sigma}\left(1 - f_{dir} + \frac{f_{dir}}{2\cos\theta}\right) + \frac{1-\alpha_g}{2\epsilon_g\sigma}S_{up} \tag{5}$$

where ϵ_g and α_g represent the emissivity and albedo of the globe, respectively. In Liljegren's original formulation, the radiation components, S_{up}, L_{down} and L_{up}, were approximated as:

$$L_{down} = \sigma \epsilon_a T_a^4 \tag{6}$$

$$L_{up} = \sigma \epsilon_{sfc} T_{sfc}^4 = \sigma T_a^4 \tag{7}$$

$$S_{up} = \alpha_{sfc} S_{down} \tag{8}$$

where ϵ_a is the emissivity of the air; T_{sfc}, ϵ_{sfc}, and α_{sfc} are, respectively, the temperature, emissivity, and albedo of the surface.

Air temperature, humidity, wind speed, and surface downward solar radiation are required as inputs for solving T_w and T_g using Liljegren's model. For details of the calculation procedure, Refer to Liljegren et al. (2008). Liljegren's model was originally written in FORTRAN and C-language programs. We rewrote it in Cython language for the implementation in Python. Find the code availability in the Acknowledgment section.

2.2. Bias Quantification Within an Idealized Context

To gain an understanding of the factors that govern biases between these formulations, bias distributions of sWBGT and ESI are first identified within an idealized context as a function of four input variables. We apply Liljegren's model in its original form to assess biases of both metrics across selected ranges of air temperature (20°C–50°C), relative humidity (5%–95%), 2 m wind speed (0.13, 0.5, 1.0, 2.0, 3.0 m/s), and surface downward solar radiation (0, 300, 500, 700, 900 W/m^2) that span ranges frequently seen in the heat stress literature.

2.3. Bias Quantification Using ERA5 Reanalysis Data

Since climate regimes spanning across the globe are diverse, biases of different magnitudes and/or signs are expected to occur over different regions. It is useful to reveal the spatial distribution of biases and identify locations where sWBGT/ESI is exceptionally biased and their applications would cause serious under- or overestimation of heat stress and downstream impacts.

ERA5 reanalysis data (Bell et al., 2020; Hersbach et al., 2018) are used to identify the bias spatial structure in this more realistic setting. Since all four radiation components (S_{down}, S_{up}, L_{down} and L_{up}) are available from the ERA5 archive, the approximations in Equations 6 and 8 are no longer necessary. The 2 m air and dew point temperature, surface pressure, 10 m wind speed, and surface downward and upwelling solar and thermal radiation on a $0.25° \times 0.25°$ grid are used to calculate WBGT at an hourly frequency for the period 1950–2019. Vapor pressure (e) is given by

$$e = RH \cdot e_s \tag{9}$$

$$e_s = \begin{cases} 6.1121 \exp\left(17.502 \dfrac{T_a - 273.15}{T_a - 32.18}\right)\left(1.0007 + 3.46 \times 10^{-6} P\right), & \text{if } T_a > 273.15 \\ 6.1115 \exp\left(22.452 \dfrac{T_a - 273.15}{T_a - 0.6}\right)\left(1.0003 + 4.18 \times 10^{-6} P\right), & \text{otherwise} \end{cases} \tag{10}$$

where e_s denotes saturation vapor pressure; RH can be calculated from dew point temperature (T_d):

$$RH = \frac{e_s(T_d)}{e_s(T_a)} \tag{11}$$

the cosine zenith angle ($\cos\theta$) is needed to project direct solar radiation from a flux through a horizontal plane (as stored in ERA5 reanalysis archive) to a flux through a plane perpendicular to the incoming solar radiation (as required by energy budget analysis) (as denoted by $\cos\theta$ term in the denominator within Equations 4 and 5).

Since model radiation components are stored as accumulated-over-time quantities (over each hourly interval in the case of ERA5 reanalysis data), the time average of $\cos\theta$ during each interval is needed. However, when the accumulation intervals encompass sunset or sunrise, the inclusion of zeros (when the sun is below the horizon) may make the time average of $\cos\theta$ too small. Being in the denominator, this too small $\cos\theta$ would lead to an overestimation of the projected direct solar radiation and consequently too high WBGT values. A simple approximate solution to this problem is taking the average $\cos\theta$ during only the sunlit part of each interval (Refer to Di Napoli et al. (2020) or Hogan and Hirahara (2016) for the calculation procedure). In Figure S1 in Supporting Information S1, we provide an example of erroneous peaks of WBGT values around sunrise or sunset introduced by using $\cos\theta$ averaged over the whole hourly interval and also show that the peaks can be removed by averaging $\cos\theta$ only during the sunlit period.

2.4. Labor Productivity Calculation

Several different labor productivity functions have been applied to assessing heat stress-induced labor reduction (Bröde et al., 2018; Dunne et al., 2013; Foster et al., 2021a, 2021b; Kjellstrom et al., 2018). It is beyond the scope of this paper to assess the relative merits of these formulations. Here, we choose the one adopted by Bröde et al. (2018) for illustrative purposes.

Our implementation is within the framework of ISO7243, an international standard providing guidance on the assessment and control of hot environments based on WBGT (ISO, 2017). ISO7243 provides WBGT reference values ($WBGT_{lim}$) corresponding to the upper limit of the prescriptive zone for different levels of metabolic heat production rates (M in Watts):

$$WBGT_{lim} = 56.7 - 11.5 log_{10}(M) + 273.15 \tag{12}$$

For WBGT exceeding the limit value, only a fraction of each hour is allowed for working in order to ensure that the physiological strain during each hour cycle can be recuperated after the rest. This fraction can be used as an estimate of labor productivity (for example, a value of 0.5 indicates a 30-min working and 30-min rest cycle, and consequently a 50% labor productivity) and calculated as follows (Bröde et al., 2018; Malchaire, 1979):

$$\text{labor productivity} = \max\left\{0; \min\left[1; \frac{WBGT_{lim,rest} - WBGT}{WBGT_{lim,rest} - WBGT_{lim}}\right]\right\} \tag{13}$$

where $WBGT_{lim,\,rest}$ denotes the limit value of WBGT under rest with a metabolic rate of 117W.

2.5. Gridded Population Data Set

We employ gridded world population data (GPWv4.11; Center for International Earth Science Information Network - CIESIN - Columbia University, 2018) with a spatial resolution of 0.25° × 0.25° for the year 2020 after adjusting to match the country total of United Nations World Population Prospects to calculate global population-weighted labor productivity.

2.6. Generalized Extreme Value Modeling

Extreme events are of special importance in the study of heat stress because of their severe impacts. For example, some studies attempt to identify extremely rare, short-term heat stress events in the past 30 years (Raymond et al., 2020; Rogers et al., 2021) and going into the future (Casanueva et al., 2020; Chen et al., 2020; Coffel et al., 2018; Im et al., 2017; Li et al., 2017; Pal & Eltahir, 2016). In order to quantify sWBGT/ESI biases in measuring extreme heat stress, we fit a generalized extreme value (GEV) model to the time series of annual maximum WBGT, sWBGT, and ESI (calculated from hourly frequency) during 1990–2019 using ERA5 reanalysis data. The 30-year return levels of all metrics are calculated and compared at each grid cell.

The cumulative distribution function of the GEV is given by:

$$F(x) = e^{-\left[1+\zeta\left(\frac{x-\mu}{\beta}\right)\right]^{-\frac{1}{\zeta}}} \tag{14}$$

where μ, β, and ζ denote the location, scale, and shape parameters, respectively. These parameters are estimated by the method of L-moments, which are less sensitive to outliers and more reliable for a smaller sample size

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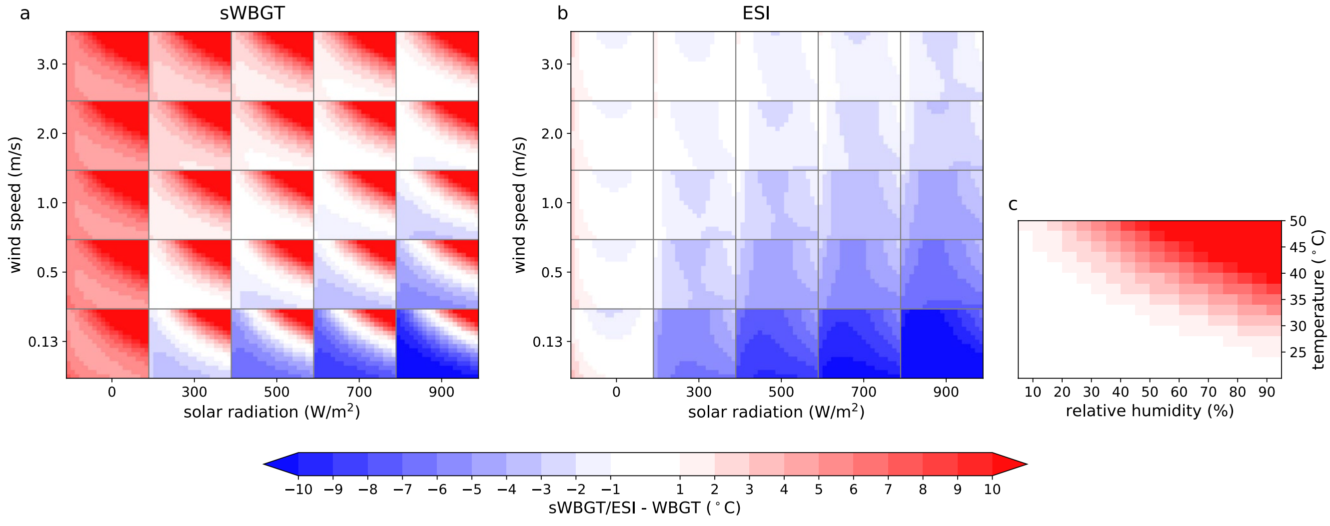

Figure 1. Bias distribution of (a) simplified WBGT and (b) environmental stress index across a selected 4-D climatic space of air temperature, relative humidity, 2 m wind speed, and surface downward solar radiation. Each small box in (a) and (b) depicts bias distribution across a range of temperature (20°C–50°C) and relative humidity (5%–95%) as shown in (c) under fixed levels of solar radiation and wind speed.

(Hosking, 1990; Hosking & Wallis, 1987). The metric values corresponding to a 30-year event are evaluated by inverting Equation 14.

The statistical significance of biases in the 30-year return level of sWBGT/ESI is evaluated by the moving block bootstrap approach (Kunsch, 1989) with a block size of three years to account for the interannual correlation. The original annual maximum time series (1990–2019) was resampled with replacement 1,000 times with sWBGT/ESI biases calculated for each replication. The 95% confidence interval of the resultant bootstrap sample is constructed using a percentile method. sWBGT/ESI biases are considered to be statistically significant (5% significance level, two-tailed test) if zero falls out of the 95% confidence interval.

2.7. Other Statistical Approaches

A two-sided *t*-test is utilized to test the statistical significance of sWBGT/ESI biases except within the GEV analysis. The Hamed and Rao Modified Mann-Kendall test (Hamed & Ramachandra Rao, 1998) is adopted for detecting any temporal trend with slopes estimated by the Theil-Sen estimator (Sen, 1968). A two-sided 5% statistical significance level is adopted throughout the paper unless otherwise stated. In addition, the calculations of all metrics, sWBGT/ESI biases, and labor productivity are performed at an hourly frequency based on which other quantities (e.g., climatological monthly averages, quantiles, annual average labor productivity, etc.) are further derived.

3. Bias Quantification

3.1. Idealized Setting

In order to understand bias structure and its dependencies on ambient conditions, we calculate sWBGT/ESI biases (sWBGT/ESI - WBGT) across ranges of air temperature, relative humidity, wind speed, and solar radiation (Figure 1). In the case of sWBGT (Figure 1a), positive biases (sWBGT > WBGT) dominate, especially during the nighttime (zero solar radiation), with bias magnitudes up to more than +10°C. This is unsurprising given that sWBGT assumes constant daylight conditions. Nevertheless, negative biases (sWBGT < WBGT) may occur under strong solar radiation and light wind conditions. Given any fixed level of solar radiation and wind speed, there tend to be larger positive biases under hotter and more humid conditions, which is a direct result of sWBGT explicitly neglecting changes in radiation and wind speed.

In comparison, ESI is mainly subject to negative biases (Figure 1b). Wind speed and solar radiation appear to be the dominant factors controlling bias magnitudes with larger negative biases under strong solar radiation and light

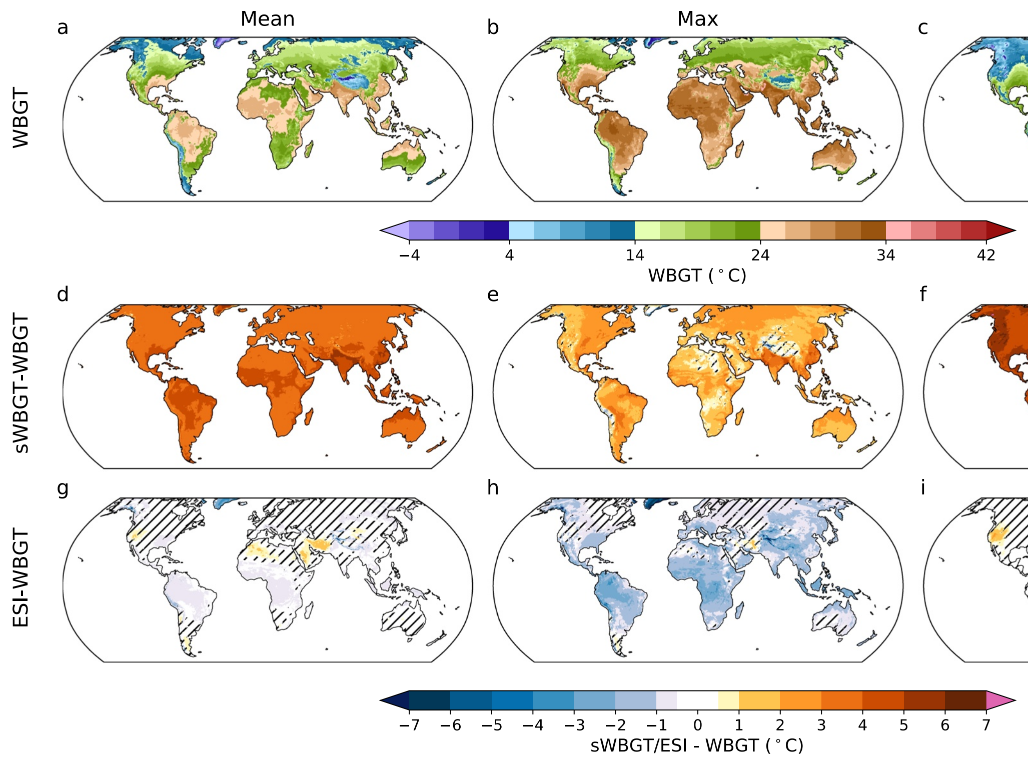

Figure 2. Climatological monthly average (CMA) of (a) daily mean, (b) maximum, and (c) minimum Wet-bulb globe temperature (WBGT) for the period 1990–2019. Biases of (d–f) simplified WBGT and (g–i) environmental stress index with respect to CMA of (d, g) daily mean (e, h) maximum, and (f, i) minimum values. Areas with biases not statistically significant at the 5% level are hatched in (d–i). Only the hottest calendar month (determined by CMA WBGT) is included.

wind (up to −10°C under 900 W/m^2 solar radiation and 0.13 m/s wind speed). Under a dry condition with relative humidity <10%, ESI exhibits relatively small negative biases (except when wind speed is low) and even positive ones during the nighttime when the bias magnitudes are overall smaller as well.

Although some combinations of the four meteorological inputs shown in Figure 1 are not necessarily physically plausible (such as large humidity and strong solar radiation) for current climate states, Figure 1 provides an overall picture of sWBGT/ESI biases across the 4-D climatic space, which can serve as a guidance for further detailed bias assessment or practical applications. For example, we expect larger overestimations by sWBGT during the nighttime (or indoor) or under a hot-humid climate such as in the tropics and larger underestimations by ESI under sunny, calm days. Next, we explore a bias structure under the more realistic setting of ERA5 reanalysis data with frequent reference to and comparison with the pattern obtained here.

3.2. Realistic Setting

3.2.1. Biases at Climatological Mean Level

ERA5 reanalysis data are applied to identifying the spatial distribution of biases within a realistic context. First, we assess biases of both metrics in terms of the climatological monthly average (1990–2019) of daily mean, maximum, and minimum values (Figure 2). Since we focus on heat stress, only the "hottest" calendar month (determined by climatological monthly average WBGT during 1990–2019) is included. A consistent and statistically significant overestimation by sWBGT is detected across most of the globe with smaller biases for daily maximum and larger biases for a daily minimum (by >4°C; Figures 2e and 2f). Areas with hot-humid summer, such as the tropics, south Asia, eastern China, and southeastern U.S., exhibit larger positive biases (>2°C for daily maximum, >5°C for a daily minimum and >4°C for daily mean; Figures 2d–2f), which is consistent with the bias structure revealed within idealized context (Figure 1a). Subtropical dry regions show smaller biases in comparison. Additionally, a topography effect is evident with smaller positive or even negative biases for daily maximum and very

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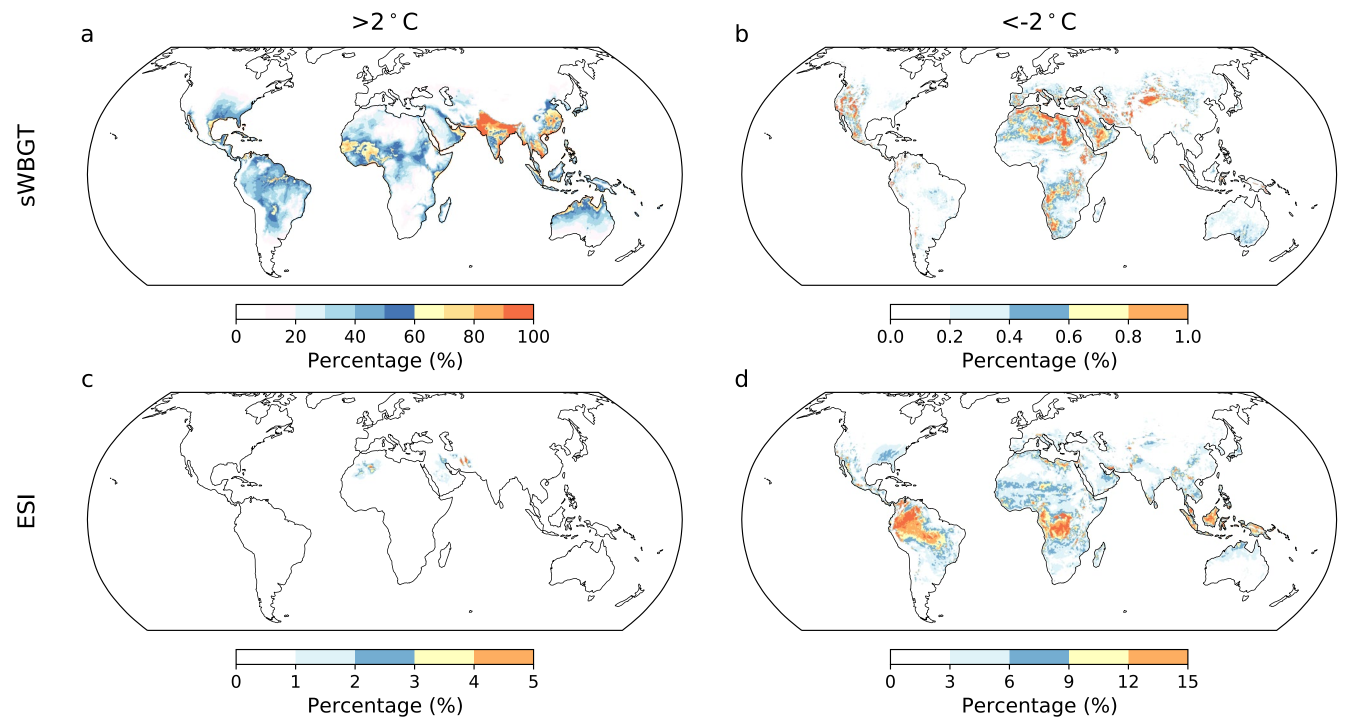

Figure 3. Occurrence percentage of (a, c) positive and (b, d) negative biases larger than ±2°C for (a, b) simplified WBGT and (c, d) environmental stress index during 1990–2019 with an additional requirement of Wet-bulb globe temperature (WBGT) exceeding 25°C. Only the hottest calendar month (defined by Climatological monthly average WBGT) is included.

strong positive biases for a daily minimum over mountainous areas such as the Himalayas and Andes (Figures 2e and 2f), although the WBGT values over these regions are generally cool (Figures 2a–2c).

ESI has smaller biases overall compared with sWBGT. Significant negative and positive biases within ±1°C occur for daily mean in the tropics and parts of subtropical dry regions, respectively (Figure 2g). Significant negative biases dominate daily maximum values except over parts of the subtropical dry area and mid-high latitudes (Figure 2h). The tropics show relatively large negative biases of −2°C to −3°C probably due to low wind speed and relatively strong solar radiation (see Figure S2 in Supporting Information S1 for associated distributions of solar radiation, wind speed, relative humidity, and air temperature conditional on daily maximum ESI) as indicated in the idealized results (Figure 1b). Subtropical dry regions, despite even stronger solar radiation (Figure S2d in Supporting Information S1), show smaller negative and even positive biases for daily maximum (Figure 2h) possibly as a result of higher wind speed and low humidity (Figures S2b and S2c in Supporting Information S1). In the case of a daily minimum, the differences between ESI and WBGT are generally small (within ±0.5°C) except for significant overestimations by 1°C–2°C over the western U.S. and the Middle East and North Africa (MENA) dry regions (Figure 2i). This agrees with the positive biases under dry nighttime conditions revealed within the idealized setting (Figure 1b).

Compared with sWBGT, ESI appears to be a better approximation particularly for the nighttime and daily mean situation. However, the larger negative biases for daily maximum (Figure 2h) imply that ESI may substantially underestimate daily peak heat stress especially when we turn from climatological mean to individual days or hours.

3.2.2. Frequencies of Relatively Large Biases

It bears mentioning that bias quantification in Figure 2 is based on 30-year climatological means, whereas bias magnitudes can be much larger over certain individual days and/or hours. Here, we count the frequencies of relatively large positive and negative biases (beyond ±2°C) based on original hourly time series during 1990–2019 for each grid cell (Figure 3), with an additional requirement of WBGT exceeding 25°C, the $WBGT_{lim}$ value for very heavy work (a metabolic rate of 520 W) according to ISO7243.

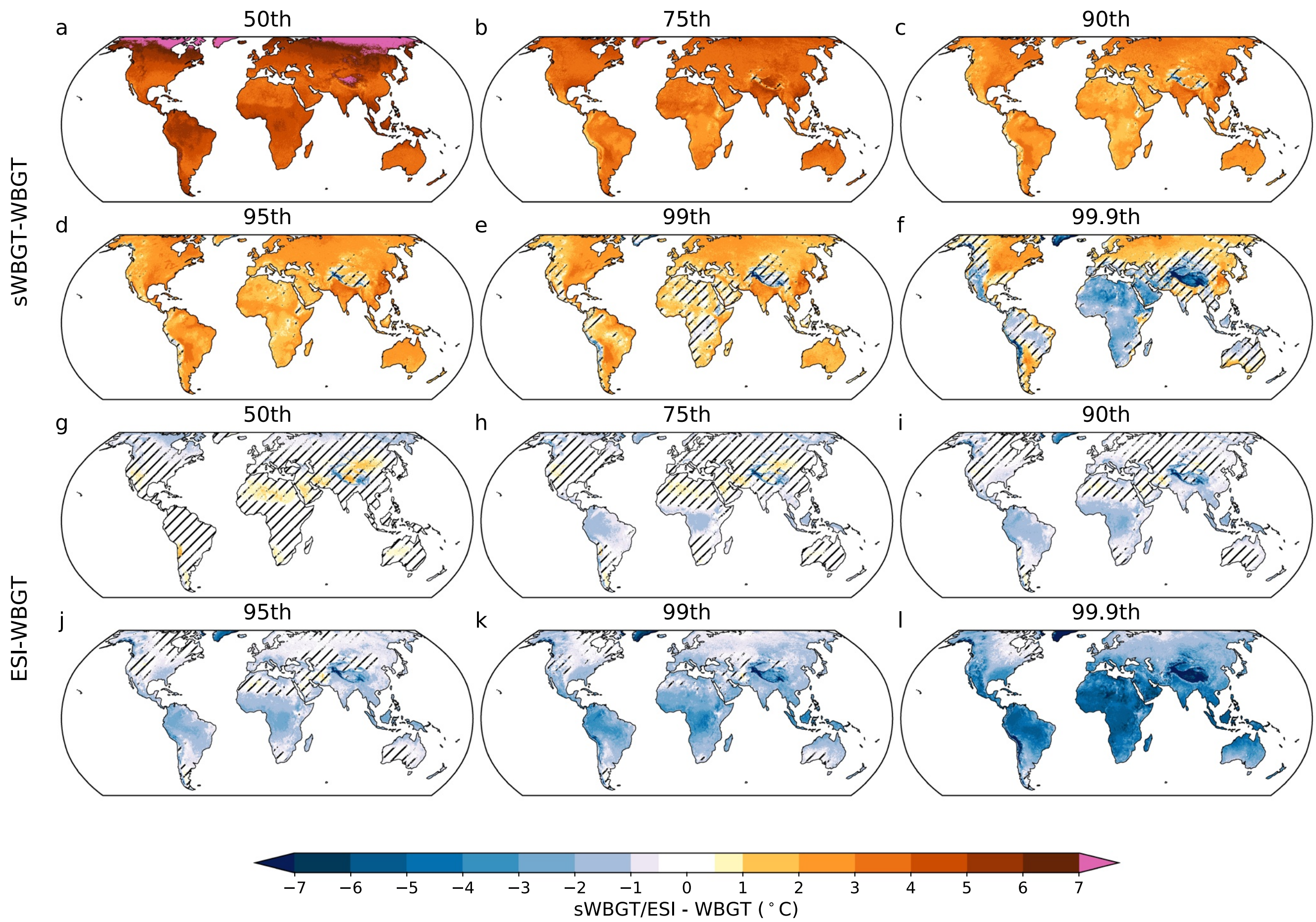

Figure 4. Biases of (a–f) simplified WBGT and (g–l) environmental stress index conditional on the (a, g) 50th, (b, h) 75th, (c, i) 90th, (d, j) 95th, (e), (k) 99th, and (f, l) 99.9th percentile values of Wet-bulb globe temperature. Areas with biases not statistically significant at the 5% level are hatched. Data of the whole year during 1990–2019 are included for analysis.

sWBGT overestimates WBGT by at least 2°C in more than 30% of cases over the tropics and other hot-humid areas and even more than 80% over the northern part of South Asia (Figure 3a). In the same region, there are still more than 50% cases even if biases magnitudes are raised to >5°C (Figure S3 in Supporting Information S1). In contrast, underestimations by more than 2°C are rare (<1%) and concentrate in subtropical dry regions (Figure 3b) presumably under dry, sunny, and calm days (Figure 1a). In the case of ESI, negative biases beyond −2°C are detected for over 10% cases in tropical areas (Figure 3d); whereas positive biases in ESI by more than 2°C are less frequent and concentrate over west Sahara and Middle East dry regions (<5%; Figure 3c).

3.2.3. Biases Conditional on WBGT Values

It is useful to know whether biases are independent of WBGT values or not. A dependency between biases and WBGTs indicates biases of different magnitudes for heat stress of different levels, among which the under- or overestimation of more severe heat stress is of particular concern. To explore it, we calculate and compare biases conditional on the 50th, 75th, 90th, 95th, 99th, and 99.9th percentile values of WBGT (Figure 4), which is done for each individual year first and then averaged across the period 1990–2019.

Both sWBGT and ESI show a clear tendency toward smaller positive or stronger negative biases moving from lower to higher percentile values of WBGT (Figure 4), suggesting a potential relationship between biases and WBGT. sWBGT conditional on the 50th percentile of WBGT shows significant positive biases beyond 3°C globally (Figure 4a), which are reduced to <2°C in the majority of the world when conditional on the 99th percentile of WBGT (Figure 4e). Significant negative biases even occur in many areas particularly over subtropical dry regions (<−2°C) when we move to the 99.9th percentile (Figure 4f). ESI exhibits small, insignificant biases (within

±1°C) worldwide conditional on the 50th percentile of WBGT (Figure 4g), which monotonically shift to strong, significant negative biases conditional on the 99.9th percentile of WBGT (<−1°C globally and −4°C to − 7°C in the low latitudes; Figure 4l). Both sWBGT and ESI exhibit large negative biases up to −10°C in high-elevation areas such as the Tibet Plateau and the Andes where nevertheless heat stress is not a concern.

The Pearson correlation coefficient r between sWBGT/ESI biases and WBGT values are also explicitly calculated. sWBGT/ESI biases and WBGT values are negatively correlated across the continental areas ($r < -0.6$ over the tropics) with the naive p-value <0.01 almost everywhere (Figures S4a–S4c in Supporting Information S1). Although the existence of autocorrelation may compromise the statistical significance level by reducing effective sample size, the smooth spatial structure in Figure S4 in Supporting Information S1 lends strong support to the validity of negative correlation. A further linear regression of sWBGT/ESI biases against WBGTs suggests a 0.2°C–0.8°C decrease of biases (toward smaller positive or stronger negative bias) per°C increase of WBGT over the majority of the land areas (Figures S4b–S4d in Supporting Information S1).

The dependence of biases on WBGT may be explained by the fact that both higher WBGT and stronger negative (or smaller positive) bias tend to be associated with strong solar radiation and light wind (Figure 1), despite a potential for higher wind speed to heat up the human body by increasing convective heat gain under extremely hot, dry environment (Foster et al., 2021a, 2021b; Havenith, 1999; Morris et al., 2019). Based on the results shown here, we expect sWBGT to largely overestimate median-level heat stress but less (or even underestimate) for more severe heat stress (such as the hottest week or 3 days of the year). ESI, in contrast, does a better job in measuring heat stress of a median level but tends to seriously underestimate those of more severity.

Here, we want to emphasize that the negative correlation between sWBGT/ESI biases and WBGT values stems from the covariation among solar radiation, wind, temperature, and humidity, which varies from region to region and potentially from now to the future. One shall not confound this correlation with a functional relationship that may be used to predict sWBGT/ESI biases from WBGT values. For instance, one may expect larger biases for both sWBGT and ESI under a warmer world with higher WBGT values, which however is not necessarily true. Since ESI bias and its correlation with WBGT values are mostly controlled by the covariation between wind speed and solar radiation (Figure 1), a warmer world with higher air temperature and negligible changes in wind speed and solar radiation would not change ESI biases in a significant way.

3.2.4. Biases of Extreme Values

The potentially strong negative biases of sWBGT and ESI conditional on higher percentile values of WBGT (Figures 4f–4l) raise a cautionary note that the most extreme heat stress may be seriously underestimated by both metrics. Here, we fit a GEV model to the time series of annual maximum WBGT, sWBGT, and ESI during 1990–2019 and estimate and compare the 30-year return levels of all metrics at each grid cell (Figure 5).

Biases of extreme values share a similar pattern with those conditional on 99.9 percentile values of WBGT (Figures 4f–4l) yet with larger magnitudes (Figures 5d and 5e). Even in extreme value sWBGT produces overestimated values in many regions (by less than 3°C in the tropics and other hot-humid areas and northern Eurasia, and by 3°C–5°C in the northeast of North America) with the notable exception of subtropical dry regions, although the overestimations are insignificant in the majority of cases. Large and significant negative biases are detected in the MENA and other regions (beyond −4°C; Figure 5d). ESI significantly underestimates WBGT by more than 3°C across most of the land surface (Figure 5e). MENA regions stand out with strong negative biases between −6°C and −10°C.

The biases structure of extreme values shown in Figure 5 is not merely a simple extension of patterns observed at climatological mean levels (Figure 2). For example, relatively small biases of ESI at a climatological mean level (Figures 2g–2i) suggest that ESI is a potentially acceptable approximation of WBGT for quantifying climatological mean heat stress. Nevertheless, ESI is expected to seriously underestimate WBGT when it comes to the most extreme heat stress conditions.

3.2.5. Local Biases in Specific Hot-Humid and Hot-Dry Regions

It is revealing to explore the bias structure in a more detailed way for two different end-member regimes relevant to heat stress, corresponding to hot-humid and hot-dry climates (Buzan & Huber, 2020). Here, we select two regions over Bangladesh (BGDH) and east Sahara (ESHR) to look into sWBGT/ESI biases in detail (Figure 6). Each region is characterized by sampling a 2° × 2° lat/lon box (Figure S5 in Supporting Information S1). In

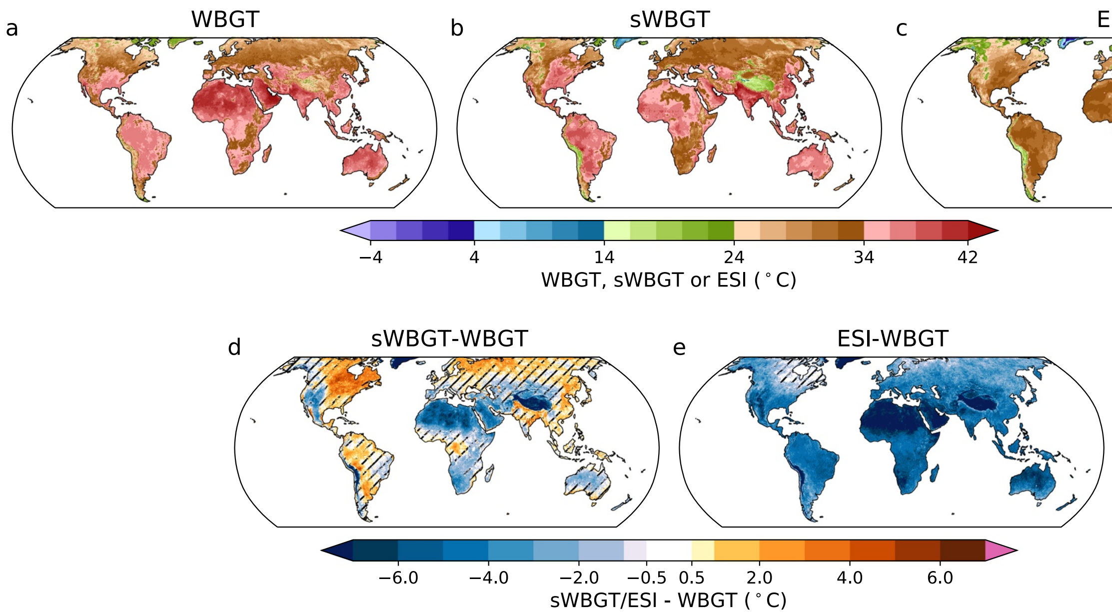

Figure 5. (a) Wet-bulb globe temperature (WBGT), (b) simplified WBGT (sWBGT), and (c) environmental stress index (ESI) return levels corresponding to a 1-in-30-year event, and their differences (sWBGT-WBGT in panel (d); ESI-WBGT in panel (e). Areas with biases not statistically significant at the 5% level are hatched in (d, e). Data of the whole year during 1990–2019 are included for analysis.

Figure 6, each column from left to right represents the sWBGT/ESI bias diurnal cycle, bias scatter plot against WBGTs during daytime and nighttime, and bias frequency heatmap including both daytime and nighttime.

sWBGT shows large positive biases (around 5°C) during the nighttime at both BGDH and ESHR (Figures 6a–6i). These positive biases decrease (even become negative for east Sahara) as moving toward the mid-day, which is consistent with previous studies (Grundstein & Cooper, 2018). sWBGT rarely underestimates WBGT in BGDH within a hot-humid climate (Figures 6a and 6b). ESHR, being hot and dry, sees positive biases in 87.25% of cases, and negative biases in 12.75% of cases nevertheless with larger magnitudes (Figure 6j). ESI shows similar diurnal cycles of biases over BGDH and ESHR with smaller biases in the nighttime, especially for BGDH (Figures 6e–6m). During the nighttime in ESHR, ESI consistently overestimates WBGT by around 2°C (Figure 6o) possibly as a result of low humidity (Figure 1b).

Consistent with the dependence of biases on WBGT values revealed in Figure 4, a significant negative correlation between daytime biases and WBGT values is identified for both metrics (Figures 6b–6n). This negative correlation is consistent with a serious underestimation of extreme heat stress by sWBGT at ESHR and by ESI at both BGDH and ESHR (Figures 6f–6n). The underestimation of extreme heat stress is especially severe at hot-dry regions for both metrics (around −10°C when WBGT is close to 40°C at ESHR) despite a mean bias close to zero or even being positive (Figures 6j–6n). Moreover, solar radiation appears to be negatively related to sWBGT biases and positively related to WBGT; wind speed, in comparison, is positively related to ESI biases and negatively related to WBGT (Figures 6b–6n). These relationships confirm the important role of solar radiation and wind speed in controlling the negative correlation between sWBGT and ESI biases and WBGTs shown in Figure S4 in Supporting Information S1. In addition, sWBGT biases and WBGT are positively related during the nighttime at BGDH (Figure 6c) probably because both sWBGT biases and WBGT values are positively associated with temperature and humidity (Figure 1a). This indicates a larger overestimation of exceptionally severe nighttime heat stress by sWBGT under a hot-humid climate.

Furthermore, hot-dry regions have a more dispersed bias distribution than hot-humid regions (compare Figure 6i, *m* and Figures 6a–6e). Bias spread is also much larger during the daytime potentially as a result of the large spatial and temporal variability in shortwave radiation.

Same calculations are also performed for another four pairs of hot-dry and hot-humid regions, and our findings remain qualitatively the same (Figures S6–S9 in Supporting Information S1).

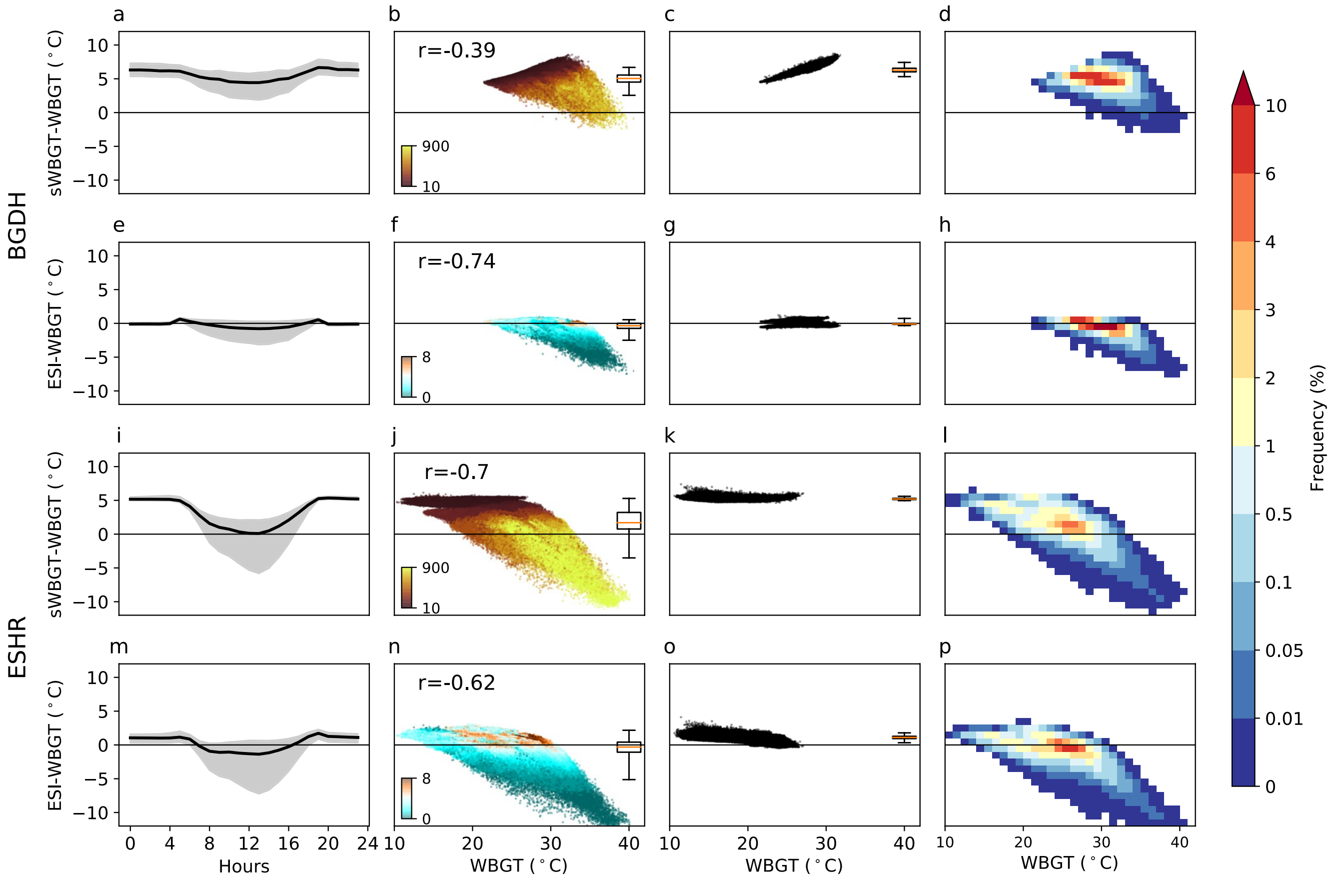

Figure 6. Biases quantification for (a–d; i–l) simplified WBGT (sWBGT) and (e–h; m–p) environmental stress index (ESI) over (a–h) Bangladesh (23°–25°N; 88°–90°E) and (i–p) east Sahara (22°–24°N; 24°–26°E). The diurnal cycle of biases is plotted in the leftmost column, with the shading area corresponding to 1st–99th percentiles. Bias scattergrams for daytime and nighttime are plotted in the middle two columns with the daytime on the left and the nighttime on the right. Boxplots placed within scattergram describe bias spread with box extending from the lower to upper quartile and whiskers representing 1st and 99th percentiles. Daytime scattergram is colored by surface downward solar radiation (W/m^2) for sWBGT bias and 10 m wind speed (m/s) for ESI bias. Pearson correlation coefficient (*r*) between daytime sWBGT/ESI biases and Wet-bulb globe temperature (WBGT) values are displayed within daytime scattergram with p-values consistently lower than 0.01. Bias frequency heatmap for both daytime and nighttime is plotted in the rightmost column. Data used to cover the period 1990–2019 with only the hottest calendar month (defined by Climatological monthly average WBGT) included.

4. Application to Labor Productivity Estimation

The sWBGT/ESI biases revealed above are expected to affect the downstream impact assessment of heat stress, which might be assessed in many ways depending on the application. Here, we take labor productivity estimation as an example to examine the impact of these biases. Labor productivity depends on working intensity measured by a metabolic rate. Here, we associate a metabolic rate of 415W with agriculture labor. Since people within the majority of industries tend to work in the daytime, and some outdoor work (such as field preparation, sowing, and crop harvesting) may rely on daylight, we assume that only daytime hours are available for working when quantifying labor productivity.

Climatological mean annual labor productivity (1990–2019) is calculated using all three metrics from ERA5 reanalysis data (Figures 7a–7c). sWBGT vastly underestimates labor productivity (as a result of overestimating heat stress) in the tropics and other hot-humid areas. The zonal average labor productivity shows large differences across the equatorial area with values unanimously above 80% according to WBGT but as low as 50% as indicated by sWBGT (Figure 7d). To put that in context that bias (30%) is comparable to labor loss in the tropics predicted for a nearly 4.0°C warming by Buzan and Huber (2020) (Figure 10 in their paper), and the predicted global labor loss from the beginning to the end of this century under RCP8.5 scenario by Dunne et al. (2013) (Figure 2 in their paper). ESI clearly did a better job with respect to deviation magnitudes. It overestimates annual labor productivity by, for example, around 10% in the tropics (Figures 7d–7f).

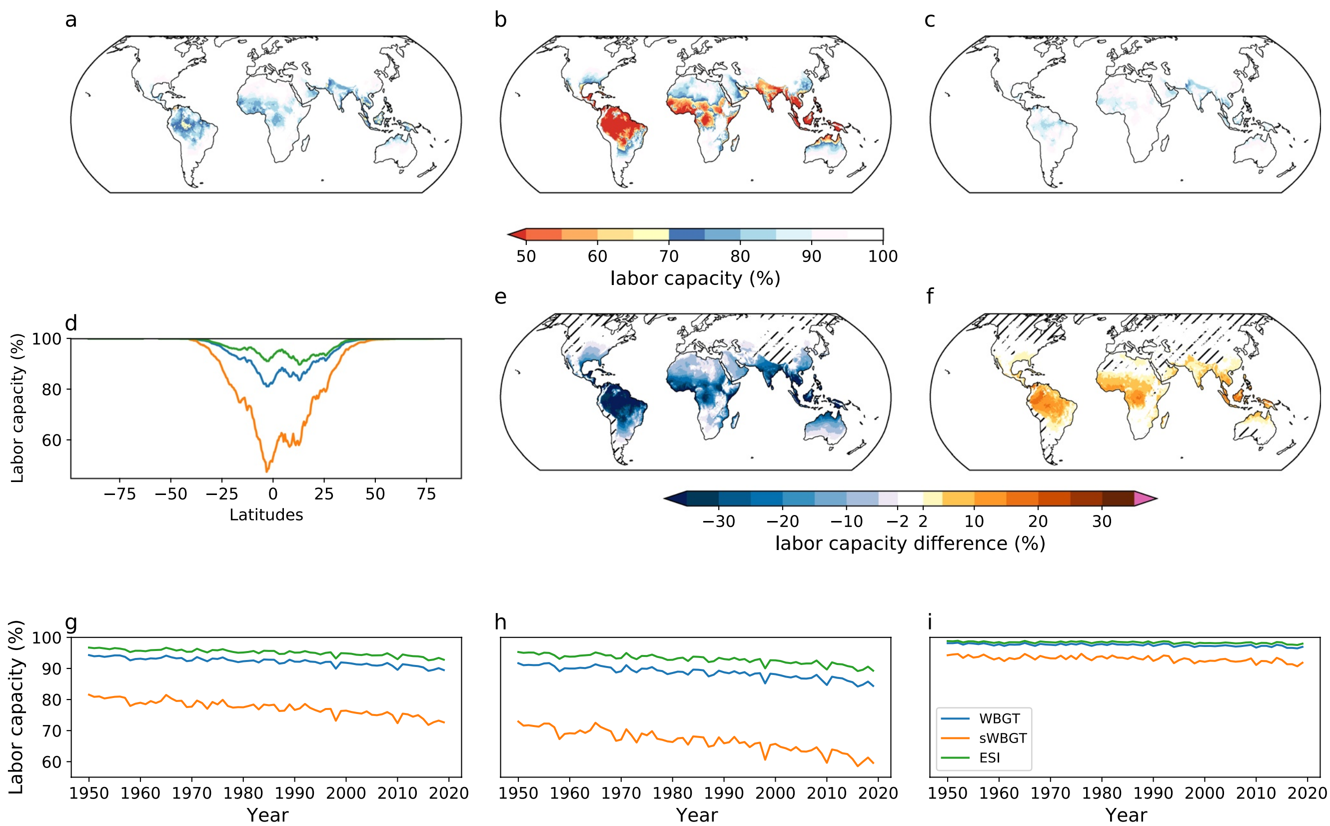

Figure 7. Annual average labor productivity for the period 1990–2019 derived from (a) Wet-bulb globe temperature (WBGT), (b) simplified WBGT (sWBGT), and (c) environmental stress index (ESI), with the zonal average value shown in (d). Labor productivity anomaly introduced by using (e) sWBGT and (f) ESI. Population weighted annual average labor productivity from 1950 to 2019 across the (g) globe, (h) low latitudes (30°S–30°N), and (i) high latitudes (outside of 30°S–30°N). Labor productivity is quantified during daytime only assuming a metabolic rate of 415W. Areas with biases not statistically significant at the 5% level are hatched in (e, f). Data during the whole year are included for the analysis.

In order to take into account population distribution and human exposure, we further calculate population-weighted average annual labor productivity for the globe, tropics, and high latitudes during 1950–2019 (Figures 7g–7i). The discrepancy between ESI and WBGT is much smaller and relatively stable along with time leading to similar decreasing trends (−0.41% and −0.54% per decade, respectively, for the global average). However, underestimation of labor productivity by sWBGT becomes increasingly large resulting in a substantially larger decreasing trends (−1.03% per decade for the global average). Namely, positive biases in sWBGT not only cause a serious underestimation of labor productivity but also an exaggeration of labor reduction tendency. This can be explained by the larger positive bias of sWBGT in the hot-humid regime (Figure 1a). Heat stress overestimation by sWBGT will be further amplified as the world warms with increasing air temperature and only small changes in relative humidity (Buzan & Huber, 2020; Byrne & O'Gorman, 2013; Byrne & O'Gorman, 2018). In contrast, wind speed and solar radiation, the main controlling factors of ESI bias, have relatively small changes with warming over land (Azorin-Molina et al., 2018; Bloomfield et al., 2021; Zelinka et al., 2013; Zeng et al., 2019).

Labor productivity in Figure 7 only includes daytime hours. However, nighttime labor is feasible for some industries like service and construction, and working hour shifting has been considered as a potential adaptation measure in the future (Takakura et al., 2018). Hence, we repeat labor productivity estimation by treating both daytime and nighttime hours as available working time (Figure S10 in Supporting Information S1). The absolute labor productivity is enhanced (Figures S10a–S10c in Supporting Information S1 vs. Figures 7a–7c). Labor productivity overestimation by ESI becomes weaker (Figure S10f in Supporting Information S1), which is consistent with the smaller bias of ESI during the nighttime (Figure 1b).

In this analysis, we have estimated annual labor productivity from hourly data, which may be not available in most archives such as CMIP and CORDEX. It is common to see studies using subdaily (Buzan & Huber, 2020; De

Lima et al., 2021), daily (Altinsoy & Yildirim, 2014; Kjellstrom et al., 2018; Liu, 2020; Orlov et al., 2020; Zhu et al., 2021), or even monthly output (Dunne et al., 2013) for a similar purpose. Although not the focus of this article, it is useful to quantify the potential error introduced thereby. Therefore, the hourly ERA5 reanalysis data are re-sampled to 8 and 4 times daily scale (calculate temporal averages of radiation flux and resample instantaneous values of other fields once each 3 and 6 hr interval), and averaged to obtain the daily mean values. The estimation of annual labor productivity (including both daytime and nighttime hours) is then repeated under each temporal resolution (Figure S11 in Supporting Information S1). The adoption of 8 times daily inputs only incurs a small error (within ±1% globally; Figure S11 in Supporting Information S1); the error introduced by 4 times daily inputs also lies between ±1% over the majority of the land surface except for an overestimation of 1%–3% in parts of the tropics (Figure S11f in Supporting Information S1). Nevertheless, labor productivity derived from daily average inputs is substantially overestimated especially in the tropics (by around 7 to >13%; Figure S11h in Supporting Information S1).

In terms of population-weighted global and annual average labor productivity, the adoption of daily average inputs introduces a consistent overestimation by around 2.2% during the period 1950–2019 (Figure S11b in Supporting Information S1). Despite this, the derived decreasing trend is similar between hourly (−0.33% per decade) and daily average inputs (−0.29% per decade). The substantial overestimation of labor productivity when using daily average inputs is not surprising since both WBGT formulation and labor productivity function are nonlinear. Particularly, all existing labor productivity functions involve a lower threshold of WBGT (e.g., 25°C for very heavy work with a metabolic rate of 520 W according to ISO7243) below which there is no labor loss. It is likely to have a daily average WBGT below this threshold but much higher WBGT values during peaking daytime hours in which case, the labor productivity estimated from daily average WBGT is too optimistic.

5. Discussion

sWBGT was soundly criticized for missing two ambient factors contributing to heat stress (Budd, 2008). However, it is widely applied because of its simplicity (Chen et al., 2020; Cooper et al., 2016; Kakamu et al., 2017; Lee & Min, 2018; Matthews et al., 2017; Schwingshackl et al., 2021; Smith et al., 2018; Willett & Sherwood, 2010). Particularly, sWBGT has been frequently adopted for estimating heat stress-induced labor productivity reduction both globally (Chavaillaz et al., 2019; Kjellstrom et al., 2009; Knittel et al., 2020) and regionally (Altinsoy & Yildirim, 2014; Liu, 2020; Zhang & Shindell, 2021; Zhu et al., 2021). For instance, under the RCP8.5 scenario labor productivity for heavy outdoor work was predicted to decrease by 38% in Southeast Asia and the Middle East by 2050 (Knittel et al., 2020) and more than 40% in South and East China by the end of this century (Liu, 2020); in the U.S., around 1.8 billion and 4.4 billion workforce hours were predicted to be lost annually by the 2050 and 2100 s under RCP8.5 scenario (Zhang & Shindell, 2021). Such estimates have been applied to informing adaptation strategies (Zhu et al., 2021) or feed into economic models for assessing the downstream socioeconomic impact (Chavaillaz et al., 2019; DARA, 2012; Zhang & Shindell, 2021). However, as we have demonstrated, the adoption of sWBGT may have introduced substantial overestimation of labor and economic loss, which may bias the design of greenhouse gas emission policy and decisions of mitigation and/or adaptation investments.

ESI has seen much less applications (De Lima et al., 2021). Its suitability depends more on the application scenarios. The predictions of different heat stress outcomes involve exposure duration of varying lengths and environmental data of different temporal resolutions (Vanos et al., 2020) and hence are affected by biases in varying degrees. For instance, monitoring exertional heatstroke in bricklayers require sub-hourly environment data and will be heavily affected by the serious underestimation of extreme heat stress by ESI during an individual peaking hour with high WBGT values. The risk estimation of classic heat stroke (Leon & Bouchama, 2015) generally uses heat stress information at a daily timescale to drive epidemiological models (Vanos et al., 2020), which therefore concerns more about biases at a daily mean level. The estimates of labor productivity reduction and the consequent economic impacts typically require heat stress information integrated over a long period such as a year. In this case, ESI may be an acceptable approximation to WBGT due to its relatively small biases under standard climatological conditions.

Except sWBGT and ESI, there are also several other approximations to WBGT being in use within heat stress literature. Orlov et al. (2020) performed a regression analysis against WBGT calculated from Liljegren's formulation and applied the resultant second-order polynomial to subsequent calculations. Some studies adopted indoor WBGT approximations ($\mathrm{WBGT_{id}} = 0.7*T_{\mathrm{pwb}} + 0.3*T_a$) to outdoors (Dunne et al., 2013; Knutson & Ploshay, 2016; Li et al., 2017; C. Li, Sun, et al., 2020; D. Li, Yuan, & Kopp, 2020; Newth & Gunasekera, 2018; Schwingshackl et al., 2021), which misses the effects of solar radiation and wind. There have been attempts to correct this missing by adding a constant (e.g., 3°C) to $\mathrm{WBGT_{id}}$ to represent the effects of solar radiation (Dasgupta et al., 2021; Kjellstrom et al., 2014; Szewczyk et al., 2021), the validity of which however has not been tested.

Here, we calculate $\mathrm{WBGT_{id}}$ using the method by Bernard and Pourmoghani (1999) as recommended in Lemke and Kjellstrom (2012) and compare it with WBGT using Liljegren's formulation under selected ranges of air temperature, relative humidity, wind speed, and solar radiation (Figure S12 in Supporting Information S1). $\mathrm{WBGT_{id}}$ differs from WBGT by varying magnitudes ranging from near zero at the nighttime to beyond −10°C at the daytime. This discrepancy is controlled by solar radiation as well as a strong interaction between solar radiation, wind speed, and air temperature, with larger discrepancy under strong radiation, low wind, and relatively lower temperature (Figure S12 in Supporting Information S1). Given large variation ranges of differences between $\mathrm{WBGT_{id}}$ and WBGT, it is unlikely that solar radiation effects can be properly captured by a constant correction factor.

We also compare annual labor productivity estimated from "$\mathrm{WBGT_{id}}$ plus 3°C" with that from Liljegren's WBGT formulation (Figure S13 in Supporting Information S1). Here, the constant correction factor is added to both daytime and nighttime hours in order to be consistent with previous studies that added 3°C to daily $\mathrm{WBGT_{id}}$ without differentiating daytime and nighttime (Dasgupta et al., 2021; Kjellstrom et al., 2014; Szewczyk et al., 2021). "$\mathrm{WBGT_{id}}$ plus 3°C" overestimates labor loss by 5 to >10% over the tropics (Figure S13 in Supporting Information S1).

In addition, many previous studies use daily (Chen et al., 2020; Schwingshackl et al., 2021) or monthly (Knutson & Ploshay, 2016; Li et al., 2017; C. Li, Sun, et al., 2020; Newth & Gunasekera, 2018) average inputs to calculate one of the approximated forms of WBGT, neglecting the fact that WBGT formulation is nonlinear involving nonlinear covariation of temperature and moisture conditions. On one hand, omission of temperature-moisture covariation makes WBGT calculated from temporally averaged inputs overestimated (Buzan et al., 2015); on the other hand, feeding daily or monthly average WBGT into a labor response function (Altinsoy & Yildirim, 2014; Chavaillaz et al., 2019; Dunne et al., 2013; Knittel et al., 2020; Liu, 2020; Orlov et al., 2020; Zhang & Shindell, 2021; Zhu et al., 2021) will result in substantial overestimation of labor productivity as we have shown above. For studies using daily average sWBGT to estimate labor productivity (Altinsoy & Yildirim, 2014; Chavaillaz et al., 2019; Knittel et al., 2020; Liu, 2020; Zhang & Shindell, 2021; Zhu et al., 2021), errors introduced by the metrics and the improper timescale may cancel each other to some extent.

For the sake of accuracy, it is recommended to use high-temporal-resolution data to calculate heat stress metrics involving the effects of multiple factors such as temperature and humidity. In addition, a heat stress module called HumanIndexMod had been incorporated into the Community Land Model (CLM), the land surface component of the Community Earth System Model since CLM4.5 (Buzan et al., 2015). HumanIndexMod can enable the calculation of several heat stress metrics (Liljegren's WBGT formulation is not included though) and thermodynamic quantities at each model time step capturing the full nonlinearity of temperature-moisture covariation. The power of Liljegren's formulation has been well recognized 10 years ago (Lemke & Kjellstrom, 2012). Climate Change Heat Impact and Prevention (Climate CHIP, 2021a; 2021b) represents one of the pioneering efforts to encourage the adoption of Liljegren's formulation for WBGT calculation (Climate CHIP, 2021a). An online WBGT calculator was developed correspondingly albeit with assumptions of indoor conditions (Climate CHIP, 2021b). However, Liljegren's formulation thus far only saw very limited applications (Casanueva et al., 2020; Jacobs et al., 2019; Orlov et al., 2019; Takakura et al., 2017, 2018). One potential reason is that Liljegren's approach is computationally intensive since it requires iterative calculations and careful treatment of latitude, date, and time of day to get solar radiation correct (Orlov et al., 2020). Moreover, Liljegren's original code was written in C and Fortran languages, which may be not familiar to most end-users. To tackle this problem, we rewrote the code in Cython, which is fast, easy to use in Python and scales well for large data set. Leveraging on parallel computing enabled by Dask, our code takes around half a minute to calculate one-year WBGT at a 3-hr frequency for a GCM with a spatial resolution of 1.5° × 1.5° using one node (24 cores) of a Linux cluster. We include Liljegren's original formulation as well as the modified version to take advantage of the full set of radiation components in climate model output. Find the details of code availability in the Acknowledgment section.

Liljegren's code is physically based and can be applied generally under many conditions including some specific working places where heat stress is of particular concern. These could include examples such as mines, aluminum smelters, and bakeries with strong radiation sources, which would invalidate the assumptions of semiempirical equations of indoor WBGT by Bernard and Pourmoghani (1999). Liljegren's formulation would still apply with straightforward modifications to the radiation part of the code (such as introducing mean radiant temperature pertaining to the working place environment).

6. Summary and Conclusions

We assessed the performance of two previously used, simple approximations to WBGT—sWBGT and ESI—against Liljegren's WBGT formulation. Bias structure across the 4-D climatic space (air temperature, solar radiation, humidity, and wind speed) was explored within an idealized context. Within this idealized framework, sWBGT is expected to overestimate WBGT during nighttime and under hot-humid conditions. Both sWBGT and ESI tend to underestimate WBGT under sunny, calm conditions. An overestimation by ESI may occur during dry nighttime. We also explored the bias distribution driven by ERA5 reanalysis data computed at hourly resolution and found results consistent with the structure revealed under idealized context. Under standard climatological conditions, we identify a substantial overestimation by sWBGT across the world and considerably smaller biases for ESI. Nevertheless, biases tend to be negatively correlated with WBGT values suggesting a serious underestimation of most extreme heat stress by both metrics, particularly in subtropical dry regions.

Given the large biases of sWBGT, we cannot recommend it as a suitable approximation to WBGT, which raises serious questions about prior work based on sWBGT. The suitability of ESI depends more on the application purposes. ESI may be acceptable for evaluating heat stress at a climatological mean level or the integrative downstream impact over a long period (such as annual labor productivity). However, the expected serious underestimation of most extreme heat stress makes ESI less suitable for epidemiological studies, extreme heat stress analysis, or as an operational index for heat warning, heatwave forecasting, or guiding activity modification at the workplace.

Liljegren's explicit formulation of WBGT should be preferred over these ad hoc approximations. With all required input variables available from climate and weather model output, we are aware of no compelling justification for using these simplifications at least in the modeling literature. Our Python implementation is straightforward to use and well suited for calculating WBGT from large-size climate model output and reanalysis data.

Notation

A	Surface area of the wick (m^2)
c_p	Specific heat of dry air at constant pressure ($J/(kg \cdot K)$)
D	Diameter of wick (m)
e_a	Ambient vapor pressure (hPa)
e_s	Saturation vapor pressure (hPa)
e_w	Vapor pressure at the surface of the wick (hPa)
ESI	Environmental stress index (°C)
f_{dir}	Fraction of the total horizontal solar irradiance due to the direct beam of the sun
h	Convective heat transfer coefficient for the wick or black globe ($W/(m^2 \cdot K)$)
L	Length of wick (m)
L_{down}	Surface downward long-wave radiation (W/m^2)
L_{up}	Surface upwelling long-wave radiation (W/m^2)
M	Metabolic heat production rate (W)
M_{Air}	Molecular weight of dry air (kg)
M_{H2O}	Molecular weight of water vapor (kg)
RH	Relative humidity
Sc	Schmidt number
$sWBGT$	simplified wet-bulb globe temperature (°C)
S_{down}	Surface downward solar radiation (W/m^2)
S_{up}	Surface upwelling solar radiation (W/m^2)

T_a	2m air temperature (K)
T_g	Black globe temperature (K)
T_{sfc}	Surface temperature (K)
T_w	Natural wet-bulb temperature (K)
T_{pwb}	Psychrometric wet-bulb temperature (K)
P	Surface pressure (hPa)
Pr	Prandtl number
WBGT	Wet-bulb globe temperature (K)
$\mathrm{WBGT_{id}}$	Indoor or shaded wet-bulb globe temperature (K)
$\mathrm{WBGT_{lim}}$	WBGT limit reference value (K)
$\mathrm{WBGT_{lim,\ rest}}$	WBGT limit reference value under resting metabolic rate (117 W; K)
α_g	Albedo of the globe
α_{sfc}	Surface albedo
α_w	Albedo of the wick
μ	Location parameter of the GEV model
β	Scale parameter of the GEV model
ζ	Shape parameter of the GEV model
r	Pearson correlation coefficient
σ	Stefan-Boltzmann constant ($W/(m^2 \cdot K^4)$)
ϵ_a	Emissivity of the atmosphere
ϵ_g	Globe emissivity
ϵ_{sfc}	Surface emissivity
ϵ_w	Emissivity of the wick
θ	Solar zenith angle (radian)
ΔF_{net}	Net radiative gain by the wick from the environment (W)
ΔH	Heat of vaporization ($J/(kg \cdot K)$)

Conflic of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Our WBGT code along with a Jupyter notebook introducing its usage is deposited in Zenodo (https://zenodo.org/record/5980536). Data analyses were performed on Purdue University's high-performance computing cluster using Python (Van Rossum & Drake, 2009) and CDO (Schulzweida, 2019). The following Python packages were utilized: Numpy (Harris et al., 2020), Scipy (Virtanen et al., 2020), Xarray (Hoyer & Hamman, 2017), Dask (Dask Development Team, 2016), Matplotlib (Hunter, 2007), Cartopy (Met Office, 2010 - 2015), and pyMannKendall (Hussain & Mahmud, 2019). Hersbach et al., (2018) was downloaded from the Copernicus Climate Change Service (C3S) Climate Data Store (https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=form). Bell et al., (2020) was downloaded from the C3S Climate Data Store (https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels-preliminary-back-extension?tab=form). The results contain modified Copernicus Climate Change Service information 2020. Neither the European Commission nor ECMWF is responsible for any use that may be made of the Copernicus information or data it contains. Center for International Earth Science Information Network - CIESIN - Columbia University (2018) was downloaded from https://sedac.ciesin.columbia.edu/data/set/gpw-v4-population-count-adjusted-to-2015-unwpp-country-totals-rev11. Liljegren's WBGT code in C language is accessible at (https://github.com/mdljts/wbgt/blob/master/src/wbgt.c).

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